

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	A Comprehensive Measure of well-being
Maximum Marks	5 Marks

Proposed Solution report: It is to develop a Machine Learning-based Human Development Index (HDI) Prediction System that predicts the HDI using key socio-economic indicators such as life expectancy, Education Index, Gross national Income (GNI) per capita, and Mean years of Schooling.

Project Overview	
Objective	The system aims to provide fast, accurate, and reliable predictions to support decision-making and research.
Scope	The project focuses on collecting and preprocessing HDI data, training a machine learning model, and deploying it through a Flask-based web application. It enables users to input development indicators and instantly obtains the predicted HDI.
Problem Statement	
Description	Traditional methods of measuring and analysing the Human development Index involve manual calculations and complex statistical analysis, which are time-consuming and prone to errors.
Impact	The proposed system reduces manual effort, improves prediction accuracy, and provides faster analysis of human development indicators.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processing specifications	Intel Core i5(or higher) CPU
Memory	RAM specifications	8 GB RAM
Storage	Disk space for dataset, models, and application	256 GB SSD (minimum)
Software		
Frameworks	Web framework	Flask
Libraries	Machine Learning & Data Processing	scikit-learn, pandas, NumPy, matplotlib, Seaborn, Pickle
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Dataset	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv